VISUALIZATION

Introduction

Outline

- What is visualization?
- Different types of visualization
- Need for visualization
- Challenges of visualization

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What is Visualization?

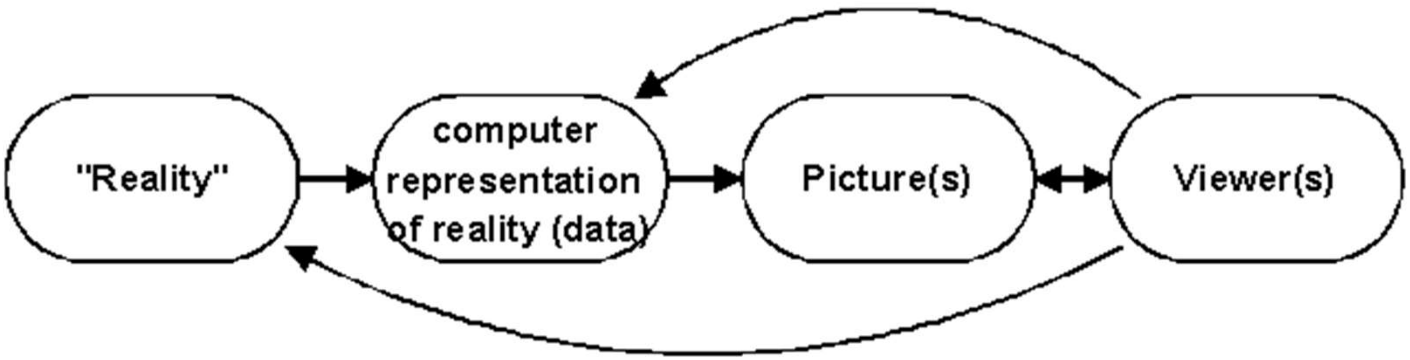
- The purpose of visualization is to convey information to people through graphical representations of data.
- Visualization refers to the use of visual representations to help people understand data and information.
- More formally, visualization is the use of interactive visual representations of data to amplify human cognition.
- A cognition -> perception process
  - Form a mental image of something
  - Internalize an understanding
- The use of computer-supported, interactive visual representations of data to amplify cognition - Card, Mackinlay Shneiderman

History of Visualization

- Visualization, in the presentation sense, is not a new phenomenon.
  - It has been used in maps, scientific drawings, and data plots for over a thousand years.
  - Most of the concepts learned in devising these images carry over in a straight forward manner to computer visualization
- Computer Graphics has from its beginning been used to study scientific problems. However, in its early days the lack of graphics power often limited its usefulness.
- With advances in the Computer Hardware the ability to create useful computer visualization has increased.
- Dozens of companies – including Google, Microsoft, IBM, Oracle and SAP – offer visualization tools.
- Thousands of companies and governments use the tools for daily operations and for longer-term strategic planning.

Basics of Visualization

- Visualization is essentially a mapping process from computer representations to perceptual representations, choosing encoding techniques to maximize human understanding and communication. The goal of a viewer might be a deeper understanding of physical phenomena or mathematical concepts, but it also might be a visual proof of computer representations derived from such an initial stage.
- “The purpose of visualization is insight, not pictures”
  - Insight: discovery, decision making, explanation

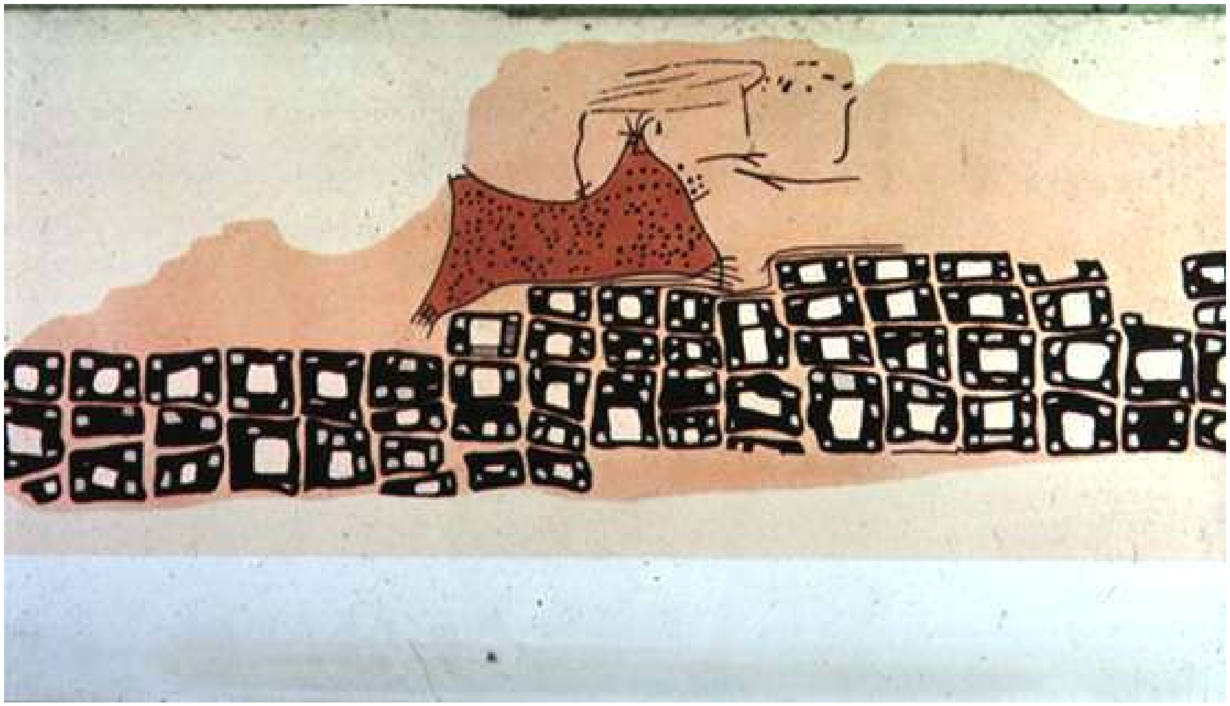


- “Visualization is really about external cognition, that is, how resources outside the mind can be used to boost the cognitive capabilities of the mind.” - Stuart Card, Xerox PARC

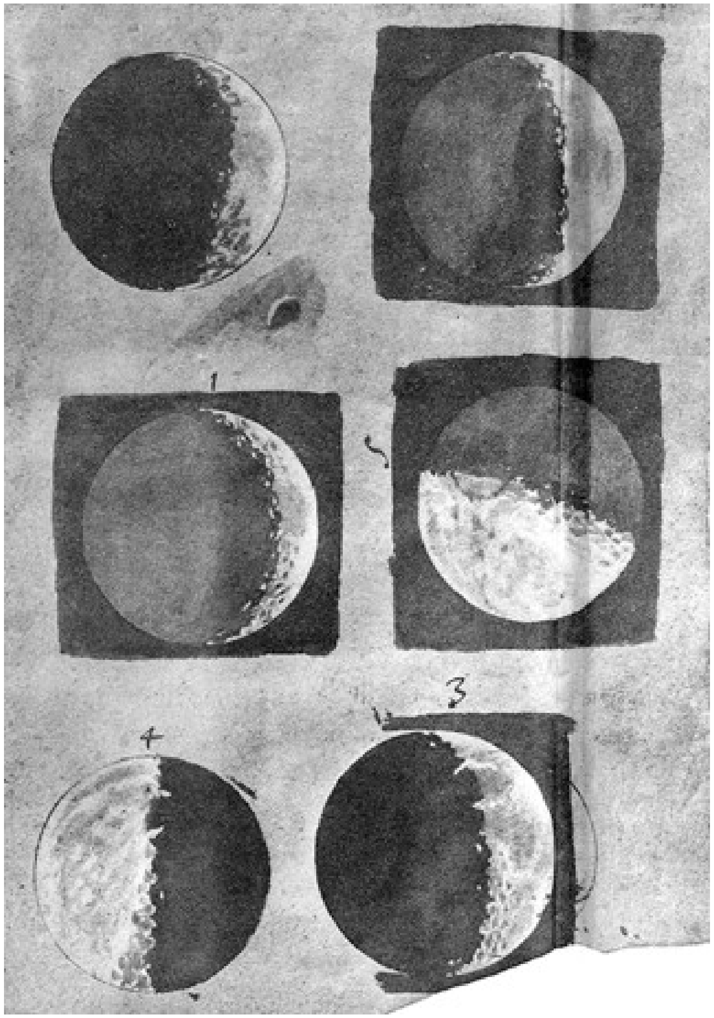
Visualization Goals

- **Record** information
  - Blueprints, photographs, seismographs, …
- **Analyze** data to support reasoning
  - Understand your data better and act upon that understanding
  - Develop and assess hypotheses (visual *exploration*)
  - Find patterns and discover errors in data
- **Presentation** – Communicate and inform others more effectively
  - Share and persuade (visual *explanation*)

Record

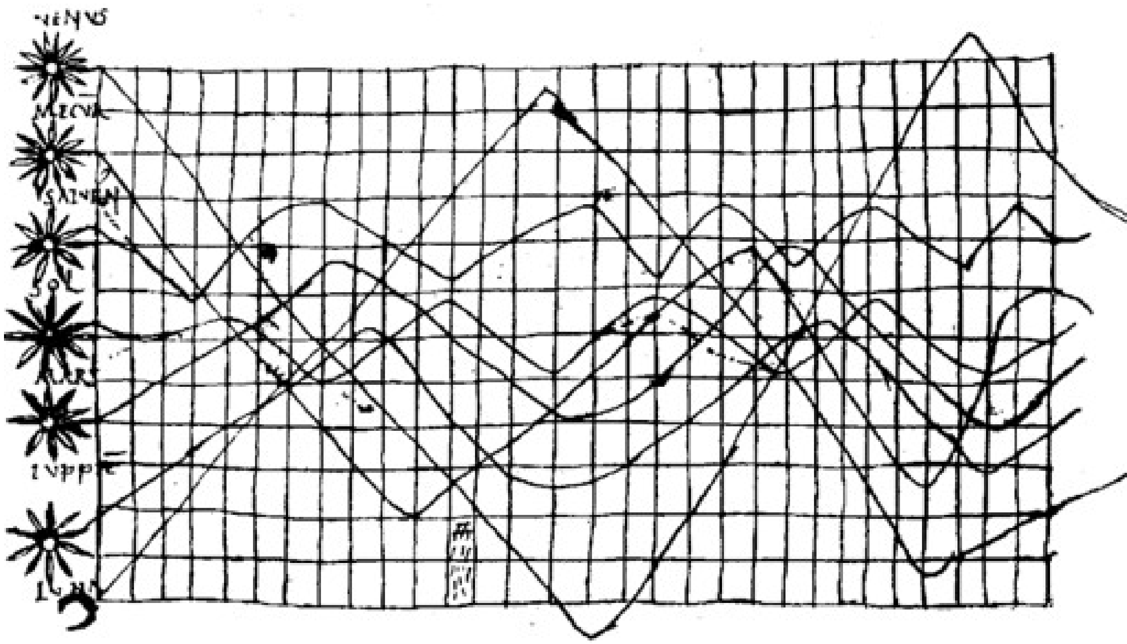


Konya town map, Turkey, c. 6200 BC

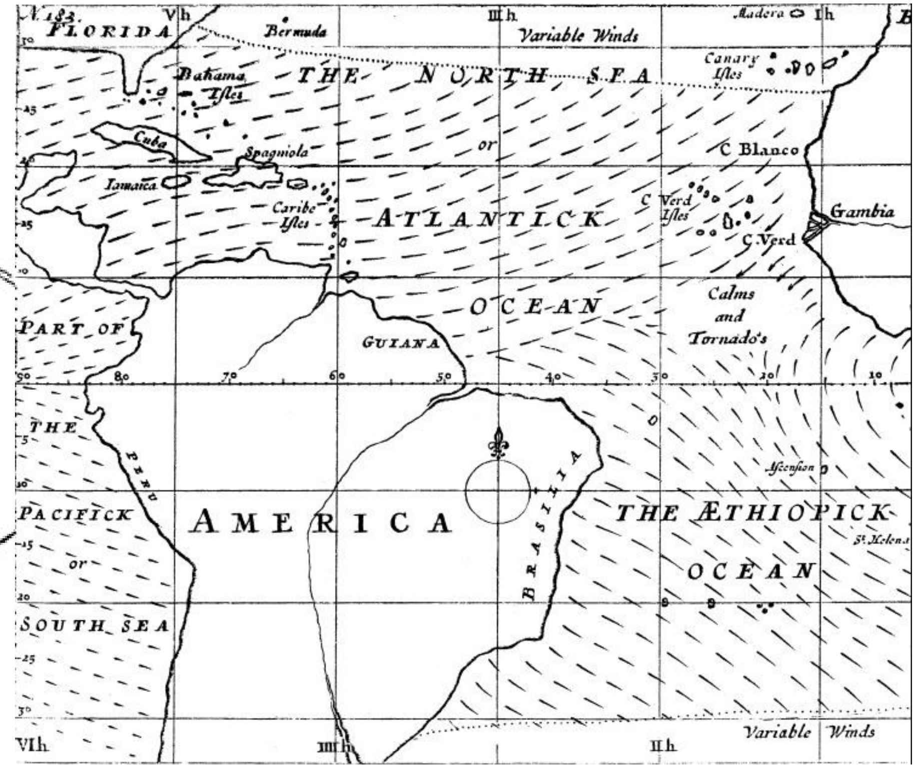


Galileo Galilei, 1616

Analyze

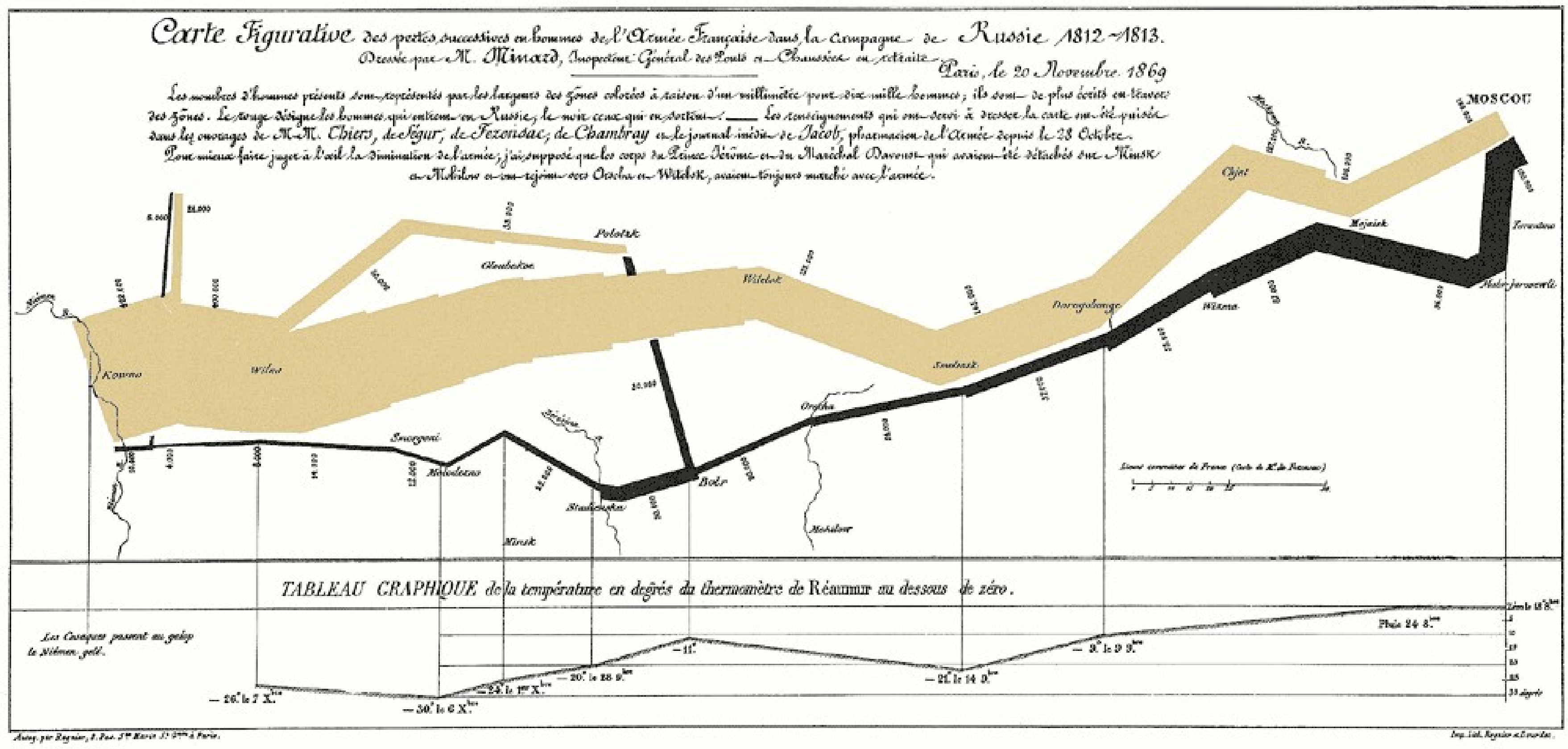


Planetary Movement Diagram, c. 950



Halley's Wind Map, 1686

Communicate



Napoleon's invasion of Russia in 1812, by Jacque Minard

- The visualization shows six types of data: the number of Napoleon's troops, distance, temperature, latitude and longitude, direction of travel, and location relative to specific dates.
- Good example to showcase the usefulness of an effective visualization – it can convey a lot of information in a single picture.
- Statistician professor *Edward Tufte* described this visualization as **what may well be the best statistical graphic ever drawn**

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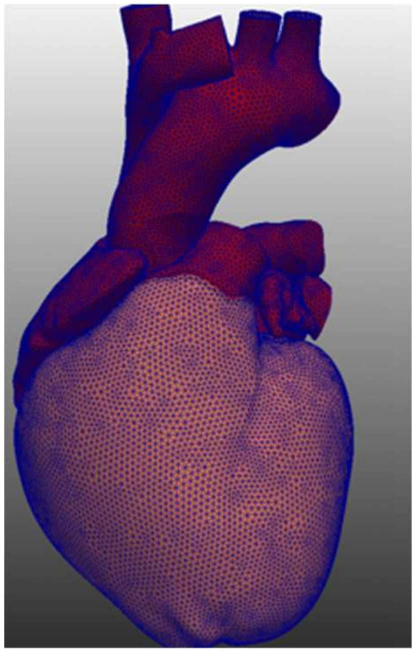
Classification of Visualization

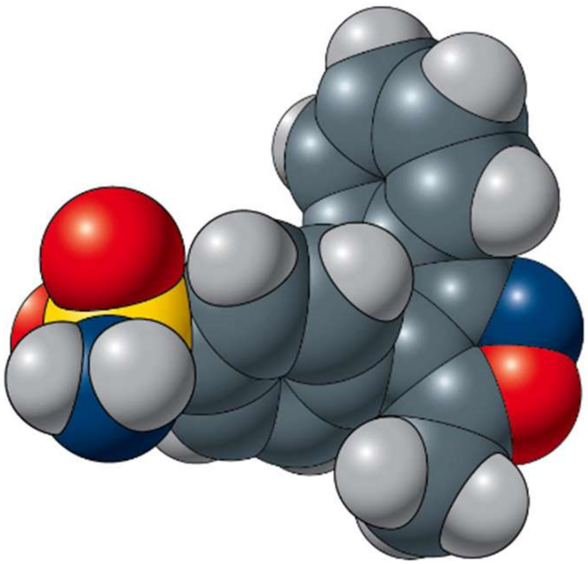
Data Visualization can be classified into 2 major areas:

- **Scientific Visualization**: Primarily relates to and represents something physical or geometric (Often 3-D)
  - Air flow over a wing
  - Stresses on a girder
  - Torrents inside a tornado
  - Organs in the human body
  - Molecular bonding
- **Information Visualization**:
  - Taking items without a direct physical correspondence and mapping them to a 2-D or 3-D physical space.
  - Giving information a visual representation that is useful for analysis and presentation

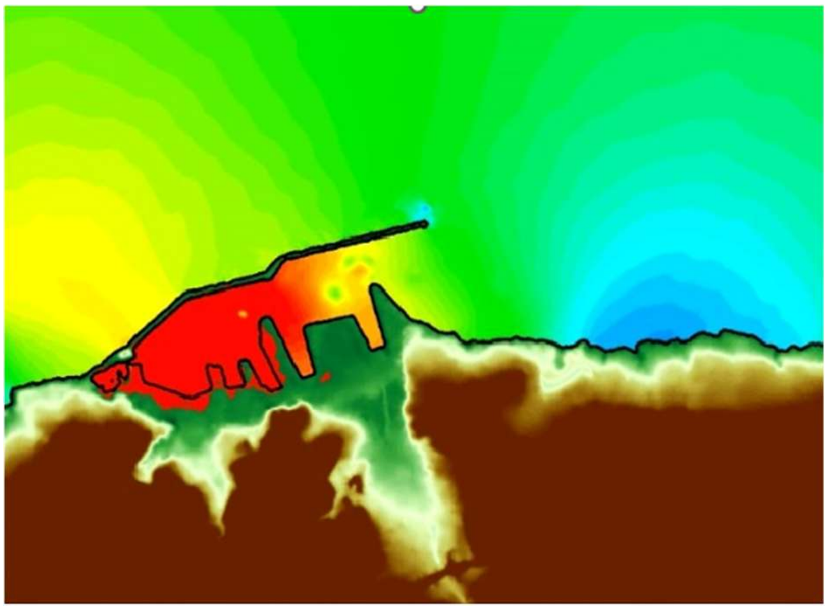
Scientific Visualization



Organs

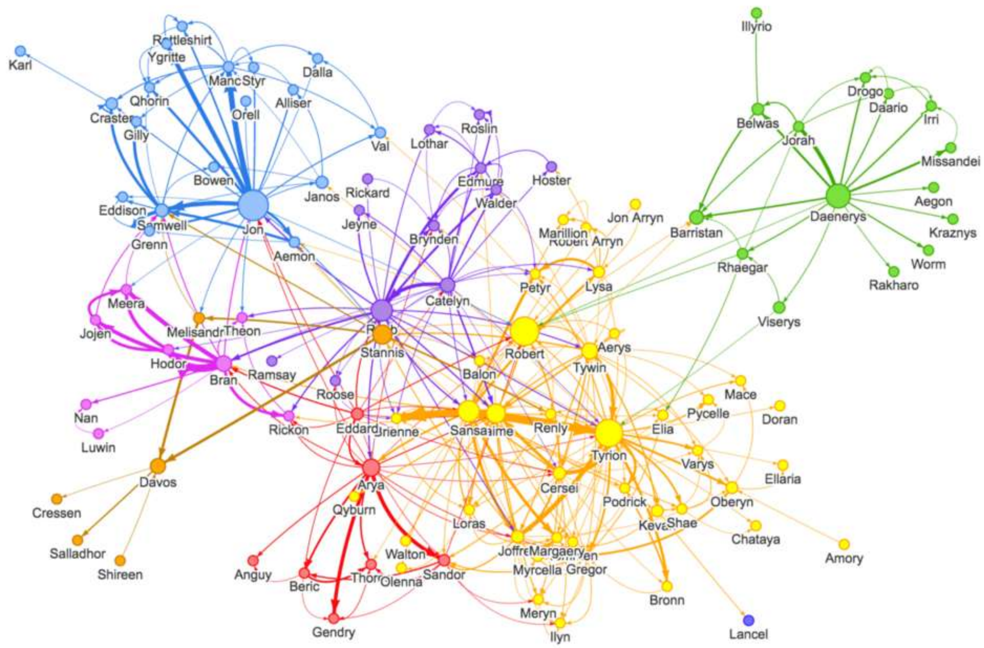


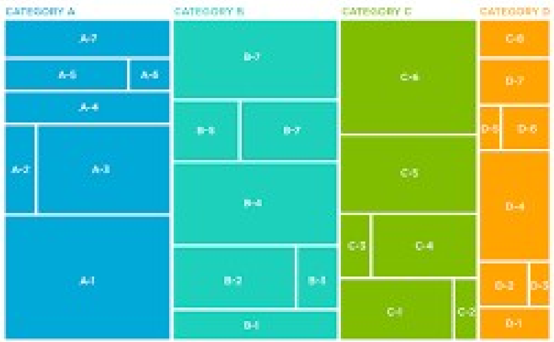
Chemical structures



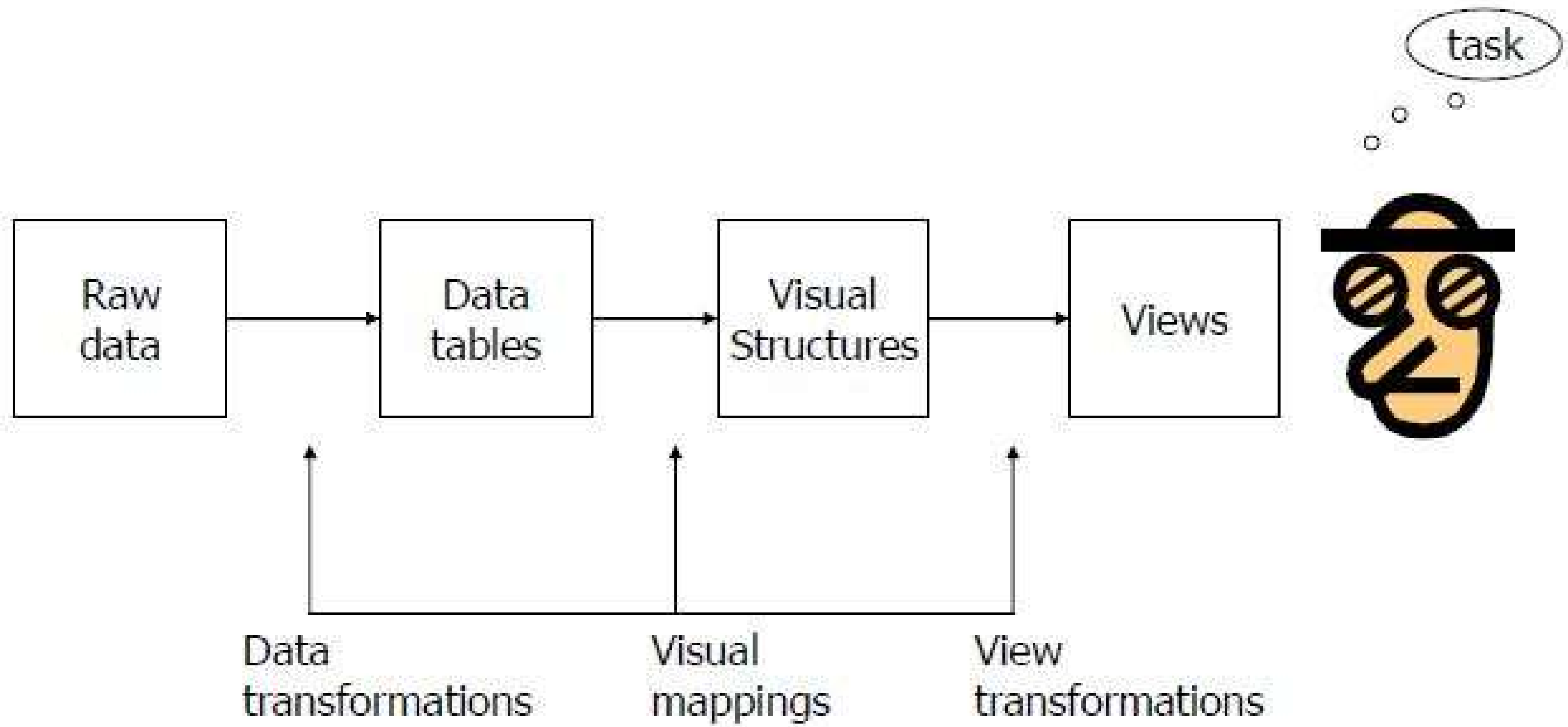
Tsunami waves

Information Visualization





Information Visualization Processing Model



"The effectiveness of information visualization hinges on two things: its ability to clearly and accurately represent information and our ability to interact with it to figure out what the information means."
S. Few, Now you see it

Related Areas

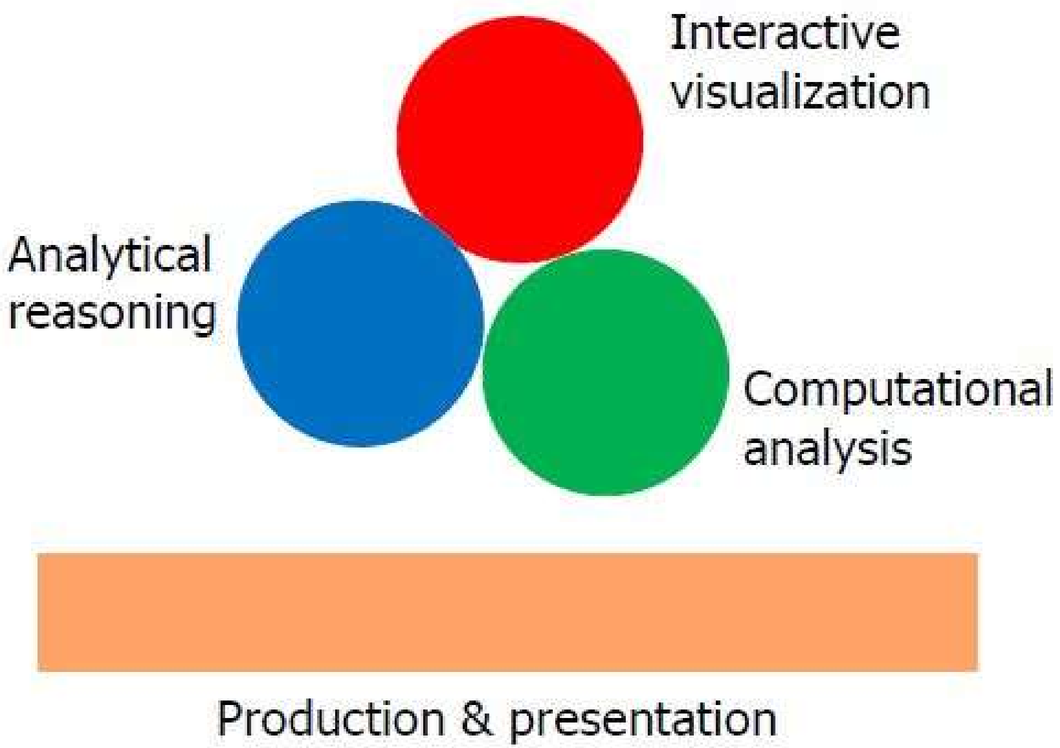
- Many other techniques for data analysis
  - Statistics, DB, Data mining, Machine learning, Information Retrieval

- Visualization is influenced by many other areas
  - Computer graphics, Human-computer interaction, Cognitive psychology, Graphic design, Cartography, Art

Visual Analytics

- Visual analytics is the science of analytical reasoning facilitated by interactive visual interfaces
- Visual analytics combines automated analysis techniques with interactive visualizations for an effective understanding, reasoning and decision making on the basis of very large and complex data sets

Visual Analytics – Main Components



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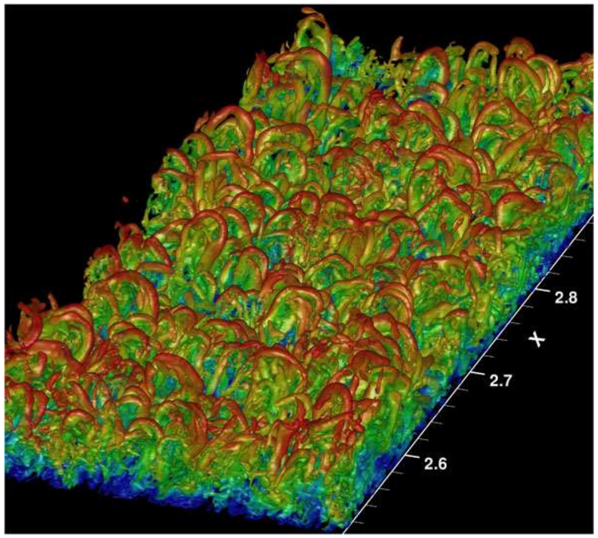
Big Data

Computers, internet and web give people access to an incredible amount of data
- news, sports, financial, purchases, etc...

New advances will add more data

Internet of Things



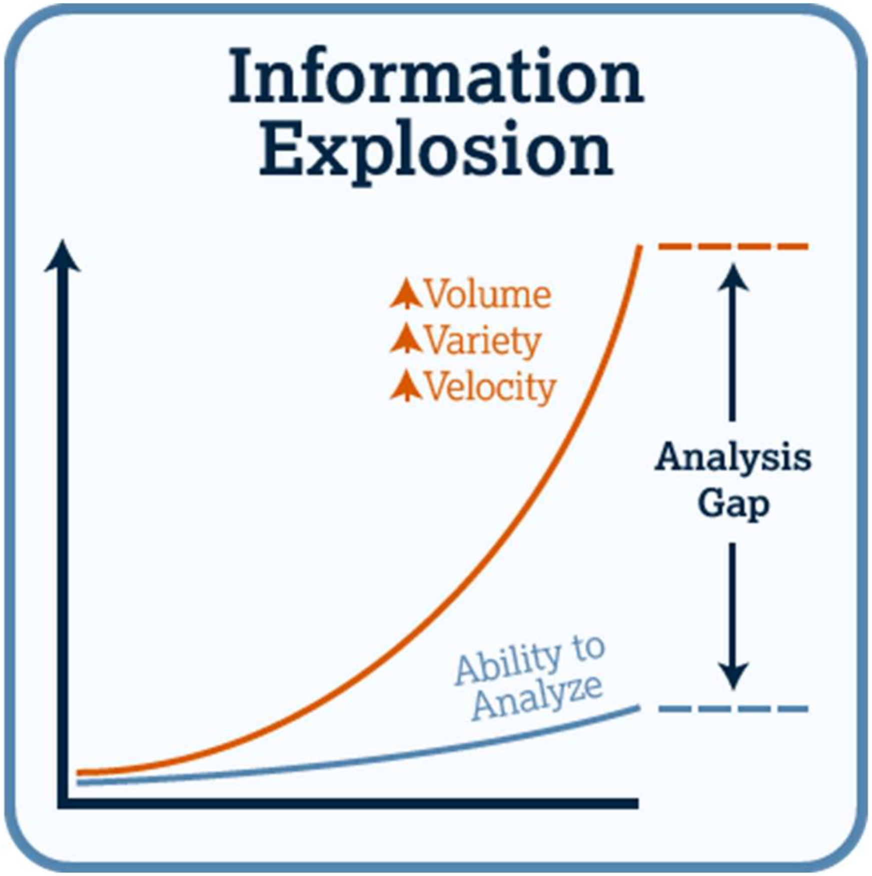
Meteorological Data

Wearable's

Genome Sequence

Information Explosion

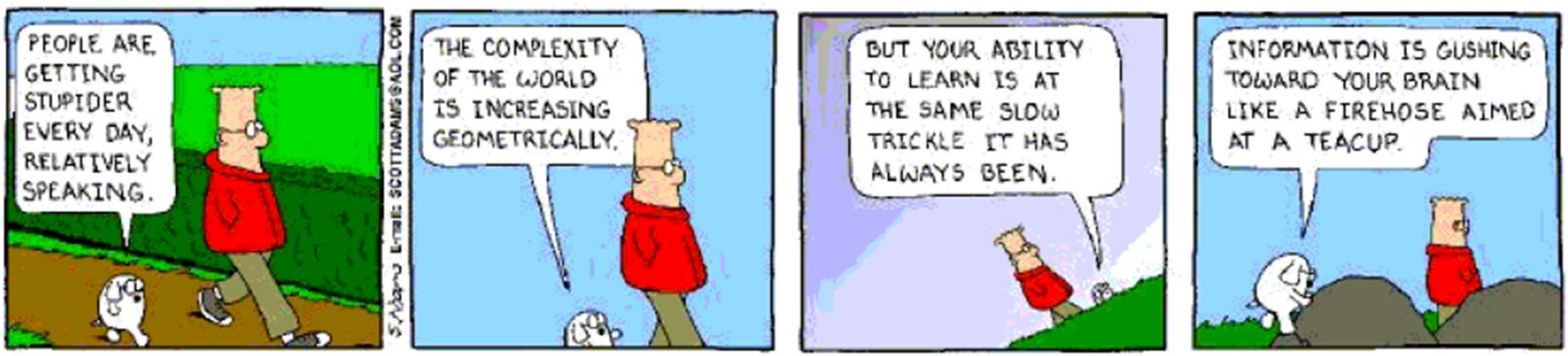
- “The ability to take data—to be able to understand it, to process it, to extract value from it, to visualize it, to communicate it—that’s going to be a hugely important skill in the next decades, … because now we really do have essentially free and ubiquitous data.”
  - Hal Varian, Google’s Chief Economist, The McKinsey Quarterly, Jan 2009
- 402.74 million terabytes of data are created each day
- It is estimated that 90% of the world's data was generated in the last two years alone.
- Given the data overload, making sense of the data and using it for decision-making is difficult



https://explodingtopics.com/blog/data-generated-per-day

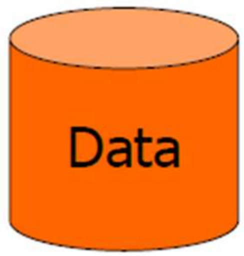
Data Overload

- How to make use of the data?
  - How do we make sense of the data?
  - How do we harness this data in decision-making processes?
  - How do we avoid being overwhelmed?



Problem

Web,
Books,
Papers,
Game scores,
Scientific data,
Biotech,
Shopping
People
Stock/finance
News



Data Transfer

How?

Vision: 100 MB/s
Ears: <100 b/s
Telepathy
Haptic/tactile
Smell
Taste

Human Vision

- Highest bandwidth sense
- Fast, parallel
- Pattern recognition
- Pre-attentive
- Extends memory and cognitive capacity (Multiplication test)
- People think visually

- **Impressive. Lets use it!**

When to apply Visualization?

- Visualization most useful in **exploratory data analysis**
  - Don't know what you're looking for
  - Don't have a priori questions
  - Want to know what questions to ask
- Search (OK)
  - Finding a specific piece of information
- Browsing (Better)
  - Look over or inspect something in a more casual manner, seek interesting information
- Visuals can frequently take the place of many words
- Visuals can summarize, aggregate, unite, explain, …
  - <u>Sometimes words are needed, however</u>

Applications of Visualization

- Political reporting and forecasting: As seen on TV and in the papers in election season.
- News reporting: visualizations used by newspapers and news TV channels
- Social science and economics data: Census and other surveys, and micro and macro economic trends.
- Social networking and web traffic: To understand patterns of communication
- Business intelligence and business dashboards: To forecast sales trends, understand competitive marketplace positions, allocate resources, manage production and logistics.
- Text analysis: To determine trends and relationships for literary analysis and for information retrieval.
- Criminal investigations: To portray the relationships between event, people, places and things.
- Performance analysis: Of computer networks and systems.
- Software engineering: Developing, debugging and maintaining software.
- Bioinformtics: To understand DNA, gene expressions, systems biology.

Advantages of Visualization

- Facilitating awareness and understanding
- Helping to raise new questions and supply answers
- Generating insights
- Telling a story and making a point

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Challenges of Information Visualization

- Evaluation:
  - All the benefits are not easily quantifiable and measured
  - How to measure and prove?
  - Evaluation is perhaps primary open research challenge for visualization
  - Return of Investment in the Business world
- Visual Representation:
  - Designing a cognitively useful spatial mapping of a dataset that is not inherently spatial
  - Accompanying the mapping by interaction techniques that allow people to intuitively explore the dataset
- Scale of data:
  - Challenge often arises when data sets become large
- Diversity:
  - Data of different types, forms, sizes

Reading

- Chapter 1